SQL Server internals and architecture guides

Applies to: SQL Server Azure SQL Database Azure SQL Managed Instance Azure Synapse Analytics SQL database in Microsoft Fabric

The following guides are available. They discuss general concepts and apply to all platforms that use the SQL Database Engine, unless stated otherwise in the respective guide.

- Concurrency, locking, and contention
- Storage engine architecture and I/O
- Query execution and optimization
- Memory, threads, and internal scheduling
- High availability, migration, and validation
- Connectivity and authentication

Concurrency, locking, and contention

Use these guides to understand how SQL Server manages concurrent access to data and internal structures, and how to diagnose contention-related issues.

Expand table

Guide	Description
Transaction locking and row versioning guide	Explains the locking and row versioning mechanisms that SQL Server uses to preserve transaction integrity. Describes how applications can efficiently control transactions.
Deadlocks guide	Deep dive on Database Engine deadlocks that competing locks cause. Explains how deadlocks form and how SQL Server detects and breaks them.
Diagnose and resolve latch contention on SQL Server	Focuses on identifying and resolving latch contention (notably page latch contention) in high-concurrency SQL Server workloads.
Diagnose and resolve spinlock contention on SQL Server	In-depth guide on identifying and resolving spinlock contention in high-concurrency SQL Server workloads.

Storage engine architecture and I/O

Use these guides to understand how SQL Server stores, accesses, and maintains data on disk.

Expand table

Guide	Description
Page and extent architecture guide	Describes page and extent structures and how pages and extents are organized within data files.
SQL Server I/O fundamentals	Explains why I/O is core to the engine and discusses efficiency articles such as drive caching principles. It also discusses I/O reliability requirements.
SQL Server transaction log architecture and management guide	Explains the transaction log's role and provides details on the physical and logical architecture of the log.
Ghost cleanup process guide	Describes ghost cleanup as a background process that physically removes rows previously marked for deletion.

Query execution and optimization

Use these guides to understand how SQL Server compiles, optimizes, and executes queries.

Expand table

Guide	Description
Query processing architecture guide	Describes how the Database Engine processes queries across storage architectures. It covers optimization and reuse through execution plan caching.
Index architecture and design guide	Covers index architecture and fundamentals. It provides best practices for designing effective indexes.

Memory, threads, and internal scheduling

Use these guides to understand how SQL Server manages memory and CPU resources internally.

Expand table

Guide	Description
Memory management architecture guide	Describes SQL Server memory architecture and how SQL Server acquires and uses memory. It includes background on OS virtual memory.
Thread and task architecture guide	Describes threading and task concepts in the context of OS scheduling. It explains how work is executed through threads and tasks.
Security cache concepts	Explains how SQL Server uses a security cache to validate permissions for principals accessing securables.

High availability, migration, and validation

Use these guides when deploying, migrating, or operating SQL Server in production environments.

Expand table

Guide	Description
Always On availability groups troubleshooting and monitoring guide	A troubleshooting and monitoring guide that also explicitly serves as a landing page pointing to other published resources for common AG scenarios and tools.
Post-migration validation and optimization guide	Frames post-migration as reconciling data accuracy and completeness and uncovering performance issues, then enumerates common post-migration performance scenarios.

Connectivity and authentication

Use this guide to understand how clients authenticate and connect to SQL Server.

Expand table

Guide	Description
Trace the network authentication process to the Database Engine	Walks through network traces that capture TCP connection establishment handshakes and authentication sequences between client and server.

Last updated on 09/21/2026